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Reflective Report

Enhancing the TaskManager with AI-Assisted Software Engineering

1. Introduction

This report reflects on the development of a task management application and the role of artificial intelligence tools in the software engineering process. The project was completed when we received it but it needed some enhancements as part of the Software Engineering program at WeThinkCode.

During my time with WeThinkCode, I gained practical software engineering experience through project-based learning. One such project involved developing a command-line task management application in Java, titled **TaskManager**.

The primary objective of the project was to enhance the functionality of a task management tool as well as being familiar with Large Language Models (LLMs) as development aids. These AI tools were used to analyse warnings, identify potential design issues, and suggest improvements to the codebase.

Although submitting a reflection was not a requirement of the project, documenting the process provided an opportunity to further analyse the development experience and evaluate the role of AI-assisted tools in software engineering practice.

This report aims to:

- Demonstrate the application of core software engineering principles within the project.
- Describe the design and implementation of the TaskManager application.
- Explain how AI tools were used to assist with identifying issues and improving the code.
- Reflect on the lessons learned during the development process.

Overall, this report highlights the ability to integrate modern AI tools into software development workflows while maintaining critical evaluation of their use.

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2. Project Overview

The **TaskManager** is a command-line application designed to assist users in organizing and managing their tasks. The system allows users to create, update, track, and analyse tasks using terminal-based commands.

The application includes several core features:

- Creating tasks with attributes such as:
    - title
    - description

    - priority level
    - due date
    - tags
- Updating existing tasks
- Marking tasks as completed
- Identifying overdue tasks
- Managing task tags
- Generating statistics related to task completion and status

While working with this project, I got to understand the importance of having **clean, maintainable, and well-structured code**, which will allow other developers to understand and improve in the future.

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3. System Architecture

The project follows a **layered architecture**, where each component of the system is responsible for a specific set of tasks. This architectural approach improves maintainability and allows different parts of the system to evolve independently.

The application consists of three primary layers:

1. **Model Layer**
2. **Application Layer**
3. **Command-Line Interface Layer**

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4. Model Layer

The **Model Layer** represents the domain of the application and contains the core data structures and logic related to task management.

This layer defines how tasks are structured and how they behave within the system. It includes several key components:

- **Task** – Represents an individual task and contains attributes such as title, description, priority, status, due date, and associated tags.
- **TaskPriority** – Defines the priority level assigned to tasks.
- **TaskStatus** – Represents the current state of a task (e.g., pending or completed).

The **Task** class serves as the central entity of the system and manages various aspects of task functionality, including:

- Task creation
- Completion tracking

- Overdue task identification
- Tag management

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5. Application Layer

The **Application Layer** contains the services responsible for coordinating task-related operations. It acts as the intermediary between the model layer and the user interface.

The key components of this layer include:

- **TaskManager**
- **Statistics**

TaskManager

The **TaskManager** class functions as the central service responsible for managing tasks within the application. Its responsibilities include:

- Creating tasks
- Updating existing tasks
- Retrieving tasks
- Tracking task status
- Managing task modifications

Statistics

The **Statistics** class analyses task data and provides useful insights, including:

- Tasks grouped by status
- Tasks grouped by priority
- Overdue tasks
- Recently completed tasks

By separating application logic from the user interface, this layer improves system maintainability and allows additional interfaces to be implemented in the future.

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6. Command-Line Interface Layer

The **Command-Line Interface (CLI) Layer** provides the mechanism through which users interact with the application.

This layer is implemented through the **TaskManagerCli** class, which is responsible for:

- Processing command-line input
- Interpreting user commands

- Calling appropriate application services
- Displaying results to the user

Separating the interface from application logic ensures that user interaction mechanisms do not interfere with the core functionality of the system.

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7. Software Engineering Practices Applied

Several key software engineering principles were applied during the development of the project.

7.1 Separation of Concerns

The system architecture clearly separates responsibilities across different layers:

- The **Model Layer** defines the data structures and domain logic.
- The **Application Layer** manages task operations.
- The **CLI Layer** handles user interaction.

This separation ensures that changes in one layer do not unnecessarily affect other parts of the system.

7.2 Thinking About Using AI

We used Large Language Models to help us with coding when we were working on the TaskManager project. I mostly used the AI to check the code, figure out what warnings meant, and find ways to make the design better.

With AI, I also got to understand the importance of installing libraries and the importance of Git when wanting to push a project from Visual Studio Code to GitHub.

I also got to understand that the command line can be used to successfully upload projects to GitHub after having installed Git.

AI tools helped with things like debugging and making the code better.